

Measure-to-measure interpolation using Transformers: formalization

Alexandre Quemy

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0.1 Foundations

Definition 1 (Tangential projector). The layer-normalization projector $P_x^\perp v = v - \langle x, v \rangle x$ onto the tangent space.

Lemma 2 (Projector identity, L1). For $\|x\| = 1$, $P_x^\perp$ is symmetric and idempotent, $P_x^\perp x = 0$, and $\langle P_x^\perp v, v \rangle = \|v\|^2 - \langle x, v \rangle^2$. [machine-checked]

Definition 3 (Geodesic distance). $d_g(x, y) = \arccos \langle x, y \rangle$, with $\cos d_g(x, y) = \langle x, y \rangle$ for unit vectors. [machine-checked]

0.2 Labeled axioms (missing Mathlib infrastructure)

Definition 4 (Wasserstein distance). W_2, W_1 with nonnegativity, symmetry, triangle inequality, the map-coupling bound, and Kantorovich-Rubinstein duality. [axiomatised]

Definition 5 (Continuity-equation flow). The solution map Φ_θ^t on points and on measures, with Lipschitz, bijective and parked-region-identity properties; the switch count. [axiomatised]

0.3 Self-contained leaves

Lemma 6 (Gate algebra and gate ODE, L2). $\frac{d}{dt} \langle x(t), \omega \rangle = (\cos R - \langle z, x \rangle)_+ (1 - \langle x, \omega \rangle^2)$ along the gated flow, with the gate active exactly on $\{d_g(z, x) > R\}$ (eq. B.4-B.5). [machine-checked] The kernel-checked equivalence $g(x) > 0 \iff d_g(z, x) > R$ corrects the printed (B.4), which states the gate is active on $\mathcal{B}_0 = B(z, R)$; consistency of Lemma B.2 requires the opposite sign (finding F1 in RESEARCH.md, confirmed independently by experiment E1).

Lemma 7 (Barycenter ODE, L6). $\frac{d}{dt} \langle x, \alpha \rangle = c (1 - \langle x, \alpha \rangle^2)$ and strict increase for $c > 0$ (eq. B.9). [machine-checked]

Lemma 8 (Geodesic gradient, L4). $\frac{d}{dt} d_g(x(t), \omega) = -(1 - \langle x, \omega \rangle^2)^{-1/2} \langle x', \omega \rangle$, exhibiting $-P_x^\perp \omega / \sqrt{1 - \langle x, \omega \rangle^2}$ as the Riemannian gradient (4.4). [machine-checked]

Lemma 9 (Separating hyperplane, L3). $d_g(\omega, x) \geq \pi/2$ and $\tau \in (0, 3\pi/8)$ give $\langle \omega, x \rangle < \cos(\pi/8 + \tau)$ (Prop 4.2 Step 1). [machine-checked]

Lemma 10 (Lyapunov sign, L5). $E = 1 - \cos \theta$ along $\theta' = -\alpha \sin \theta$ has $\dot{E} = -\alpha \sin^2 \theta \leq 0$ (Ex. 6.1). [machine-checked]

Lemma 11 (Ball-chain retention, L9). Per-step retention $(1 - \varepsilon)$ gives $a_K \geq (1 - \varepsilon)^K a_0$ (Lemma B.1 induction). [machine-checked]

Lemma 12 (Pigeonhole, L10). A nonempty open ball contains a point $\neq a$, so no map is constant on it (Lemma 3.4 Part 1). [machine-checked]

Lemma 13 (Linearized OT bound, L7). $W_2(T_\#^1 \mu, T_\#^2 \mu) \leq \|T^1 - T^2\|_{L^2(\mu)}$ (Lemma 5.2). [axiomatised]

Lemma 14 (Markov bound, L8). $W_2(\mu, \delta_{x_0}) \leq \eta_2 \Rightarrow 1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ (Claim 2). [informal]

0.4 Mass transport and clustering

Lemma 15 (Single ball pair, B.2). *Mass $\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1 - \varepsilon)\mu_0(\mathcal{B}_0)$. [informal] Caveat (F1): the printed parameters $U = -z\mathbf{1}^\top$, $b = \cos(R)\mathbf{1}$ make the gate active on the complement of $\mathcal{B}_0$, so the lemma transports nothing as written; the sign must be flipped to $U = +z\mathbf{1}^\top$, $b = -\cos(R)\mathbf{1}$. True once corrected.*

Lemma 16 (Ball chain, B.1). *Mass through K overlapping balls $\geq (1 - \varepsilon)^K$. [informal]*

Proposition 17 (Clustering to a point, 2.1). *A single-hemisphere measure clusters to a point; $\text{diam} \rightarrow 0$ at rate $O(\log 1/\varepsilon)$. [informal]*

Proposition 18 (Clustering to discrete, 2.2). *Disjoint-support atomless measures cluster to M atoms. [informal]*

0.5 Disentangling supports

Lemma 19 (Transport to the orthant, 3.2). *One switch moves measures into $\mathbb{Q}_1^{d-1}$. [informal]*

Lemma 20 (Disentangle non-colinear, 3.3). *Shrink a measure's hull to its barycenter direction. [informal]*

Lemma 21 (Make non-colinear, 3.4). *Two measures made non-colinear. [informal]*

Proposition 22 (Disentanglement, 3.1). *A piecewise-constant θ renders the N supports pairwise disjoint, $O(dN)$ switches. [open] Caveat (F2): the induction asserts "disjoint geodesic hulls in $\mathbb{Q}_1^{d-1} \Rightarrow$ pairwise non-colinear barycenters" without proof; the case split rests on it. Needs a supporting lemma before a faithful formalization (see `RESEARCH.md`).*

0.6 Matching discrete measures

Proposition 23 (Steer one point, 4.2). *$x_0^M \rightarrow y^M$ while inactive points stay fixed, ≤ 6 switches. [informal]*

Proposition 24 (Match ensembles, 4.1). *M points $x_0^i \rightarrow y^i$ exactly, $\leq 6M$ switches. [informal]*

0.7 Main results

Lemma 25 (Transport maps preserved, 5.1). *Existence of ψ . [informal]*

Lemma 26 (L^2 approximation, 5.4). *Approximate any ψ by a flow map. [informal]*

Theorem 27 (Theorem 1.2). *A single Transformer matches N arbitrary inputs to N arbitrary (transport-matchable) targets in W_2 . [open]*

Theorem 28 (Theorem 1.1). *Dirac targets: matching with $O(dN)$ switches and norm $O(dN/T + \log 1/\varepsilon)$. [open]*

Proposition 29 (Generic $O(1)$ switches, 6.2). *For generic i.i.d. discrete inputs a single constant parameter disentangles almost surely. [informal]*